

## PROJECT FLOW

### Problem Identification and Project Understanding

Story 1: Understand and clearly define the agricultural problem that the crop recommendation system is designed to solve.

Story 2: Collect and study business requirements to determine the project's goals, scope, and expected outcomes.

Story 3: Review existing research papers and machine learning techniques related to crop recommendation systems.

Story 4: Assess the business value and social benefits of implementing an AI-based crop recommendation solution.

### Data Collection and Exploration

Story 1: Obtain a reliable agricultural dataset from trusted sources.

Story 2: Install and import the required Python libraries for data analysis, visualization, and machine learning.

Story 3: Load the dataset and examine its features, structure, and target variable.

Story 4: Perform univariate analysis to study the distribution of individual variables.

Story 5: Conduct bivariate analysis to understand relationships between agricultural features and crop suitability.

Story 6: Perform multivariate analysis to identify patterns and interactions among multiple variables.

### Data Preparation

Story 1: Identify and handle missing values to improve data quality.

Story 2: Detect and treat outliers that may negatively affect model accuracy.

Story 3: Create and prepare seasonal crop-related features for model development.

Story 4: Divide the dataset into training and testing sets for model training and evaluation.

### Machine Learning Model Development

Story 1: Apply the K-Means clustering algorithm to group similar agricultural conditions.

Story 2: Train a Logistic Regression model to recommend suitable crops based on environmental factors.

Story 3: Measure and compare model performance using appropriate evaluation metrics.

Story 4: Save the best-performing model and use it to generate crop recommendations from user inputs.

### Application Development

Story 1: Develop an interactive and user-friendly web interface using HTML.

Story 2: Build the Python backend and integrate it with the trained machine learning model.

Story 3: Test and validate the complete application to ensure accurate crop recommendations and smooth performance.